

An Investigation of Priority Inversion and its Solutions in Real-Time Operating Systems.

Real-time systems are designed to perform tasks within strict timing constraints, but they can be vulnerable to security threats like covert timing channels and side-channel attacks. These attacks can exploit the system's predictable execution behavior to infer sensitive information or disrupt critical operations. Priority inversion, where a high-priority task is delayed by a lower-priority one, can exacerbate these vulnerabilities.

This research paper will explore various techniques to mitigate these security risks in real-time systems. It will begin with an examination of priority inversion, including its causes and effects on real-time scheduling and security. Then, it will analyze different strategies to address these challenges. This analysis will include the TimeDice protocol, which introduces randomization into time-partition schedules to disrupt covert timing channels, and the FT-PIP protocol, which incorporates flush tasks and priority inheritance to reduce information leakage in multiprocessor real-time systems. Additionally, the paper will investigate the use of multimode scheduling to counteract schedule-based attacks and randomized scheduling strategies like REORDER++ to defend against side-channel attacks. By comparing these approaches, the paper will provide a comprehensive overview of the current state of research in securing real-time systems, highlighting the ongoing efforts to balance timing constraints with the need to mitigate security vulnerabilities.

References

1. M. -K. Yoon, J. -E. Kim, R. Bradford and Z. Shao, "TimeDice: Schedulability-Preserving Priority Inversion for Mitigating Covert Timing Channels Between Real-time Partitions," *2022 52nd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, Baltimore, MD, USA, 2022, pp. 453-465, doi: 10.1109/DSN53405.2022.00052.
2. H. Baek and J. Lee, "FT-PIP: Flush Task Incorporated Priority-Inheritance Protocol to Reduce Information Leakage on Multiprocessor Real-Time Systems," in *IEEE Access*, vol. 9, pp. 81882-81894, 2021, doi: 10.1109/ACCESS.2021.3086304.
3. J. Ren *et al.*, "REORDER++: Enhanced Randomized Real-Time Scheduling Strategy Against Side-Channel Attacks," in *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 6, pp. 3253-3266, Nov.-Dec. 2023, doi: 10.1109/TNSE.2023.3254653.
4. J. Ren *et al.*, "Multimode Security-Aware Real-Time Scheduling on Multiprocessors," in *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 43, no. 11, pp. 3407-3418, Nov. 2024, doi: 10.1109/TCAD.2024.3445260.

5. Y. Tang, X. Jiang, N. Guan, S. Liu, X. Luo and W. Yi, "Optimizing End-to-End Latency of Sporadic Cause-Effect Chains Using Priority Inheritance," *2023 IEEE Real-Time Systems Symposium (RTSS)*, Taipei, Taiwan, 2023, pp. 411-422, doi: 10.1109/RTSS59052.2023.00042.
6. C. Sakalis, M. Sjölander and S. Kaxiras, "Seeds of SEED: Preventing Priority Inversion in Instruction Scheduling to Disrupt Speculative Interference," *2021 International Symposium on Secure and Private Execution Environment Design (SEED)*, Washington, DC, USA, 2021, pp. 101-107, doi: 10.1109/SEED51797.2021.00022.
7. J. B. Goodenough and L. Sha, "The priority ceiling protocol: A method for minimizing the blocking of high priority Ada tasks," in *Proceedings of the Second International Workshop on Real-Time Ada Issues*, Moretonhampstead, Devon, England, 1988, pp. 20–31, doi: 10.1145/58612.59371.
8. L. Sha, R. Rajkumar and J. P. Lehoczky, "Priority inheritance protocols: an approach to real-time synchronization," in *IEEE Transactions on Computers*, vol. 39, no. 9, pp. 1175-1185, Sept. 1990, doi: 10.1109/12.57058.
9. S. Davari and L. Sha, 'Sources of unbounded priority inversions in real-time systems and a comparative study of possible solutions', *SIGOPS Oper. Syst. Rev.*, vol. 26, no. 2, pp. 110–120, Apr. 1992, doi: 10.1145/142111.142126.